AI Driven Path Finder for Autonomous Vehicles

A PROJECT REPORT ON 6CS1991 Design Project-II

Submitted by

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ABSTRACT

In controlled environments such as factories, shipyard, airports use many vehicles for various tasks. All the vehicles operating in such environment are non-autonomous and require constant human interaction, so there is a need for autonomous vehicles. Autonomous navigation technologies have the potential to improve efficiency, safety and coordination of vehicles operating in controlled environments. However completely changing all the vehicles to autonomous vehicles is costly and not feasible. So, our project presents a proof-of-concept software framework for AI driven path planning designed for controlled environments. The proposed framework focuses on simulation-based proof-of-concept in graph-based environment with modular system design, while A* algorithm is used for calculating optimal and collision-free paths between defined start and goal positions. The system is implemented and evaluated using simulation experiments and a visualization dashboard that visualizes the path planning behaviour along with dynamic replanning. This simulation results demonstrate that classical path planning techniques can effectively support autonomous navigation in controlled environments with low computational overhead. The proposed framework establishes a scalable foundation for future hardware-based autonomous vehicle systems.

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LIST OF SYMBOLS, ABBREVIATIONS & NOMENCLATURE

Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

RL – Reinforcement Learning

CNN – Convolutional Neural Networks

UI – User Interface

GPS – Global Positioning System

AV – Autonomous Vehicle

Symbols

S – Start Node

G – Goal Node

O – Obstacle

H(n) - Heuristic function

G(n) - Path cost from start to node n

F(n) = G(n) + h(n) - Evaluation function

D - Distance

T – Time

1. INTRODUCTION

Various types of vehicles are employed in material handling, transportation, and logistics in most controlled industrial settings like warehouses, factories, shipyards, ports, and airports. The drivers of these vehicles are usually human drivers, and this makes the entire system reliant on the supply of labour, human decision making and manual coordination of traffic. This strategy causes the escalating labour costs, delays due to human error, inefficient movement of vehicles, and safety issues as the size of operations grows. As the technology of autonomous driving evolves, fully driverless cars can help to remove these problems. Nevertheless, most organizations cannot substitute all their current non autonomous vehicles with autonomous ones. Autonomous vehicles are expensive and need special infrastructure due to a special infrastructure, and at a large scale deploying an entire system is not feasible. Consequently, there is an increasing demand to have a more cost-effective and flexible solution that would upgrade the existing cars, instead of substituting the cars.

This project fulfils that requirement by developing a proof-of-concept low-cost hardware and software extension that would be operating in the non-autonomous vehicles which are already in use. The objective of the system is to offer autonomous functions which consist of obstacle detection, environmental perception, path planning and safe navigation without significant alterations in the actual mechanical or control systems of a vehicle. The solution is developed to be used in controlled environments where road regulations, speed limits, and layouts of the environment are predictable and thus it is easier than having full public-road autonomy. The system proposed has a sensor module, which is used to provide real-time perception, an onboard processing unit, and actuation interfaces to regulate the movement of the vehicle. On software part, computer vision is applied in the obstacle and moving objects detection, and AI based path planning algorithms create efficient and collision-free paths. To ascertain safety, reliability and performance of the system, it will be tested and trained in simulated and small-scale physical environment and then deployed to real vehicles.

Overall, this project proposes a proof-of-concept solution to the practical and scalable implementation of autonomous vehicle integration in controlled environments. The solution does not replace the current vehicles but increases their efficiency enabling organizations to reduce human efforts, increase their efficiency, and minimize errors that result in accidents and still maintaining a minimal cost of implementation.

2. LITERATURE REVIEW

2.1 Trustworthy Internet of Vehicles

This work by Jing Yang et al focuses on safe, secure and collaborative autonomous driving. It shows the potential of the Internet of Vehicle to reduce on-board system costs in autonomous vehicles through shared intelligence is considerable. However, it still faces significant challenges, including concerns over data breaches and privacy, inefficiencies and limited fault tolerance in centralized management, and the challenge of ensuring data accuracy.

2.2 Driving information logger with in-vehicle communication for autonomous vehicle research

This work by Sung et al focuses on information logger with in-vehicle communication for autonomous vehicle research. The authors describe in detail the driving information logging used in developing an autonomous vehicle. The recorded vehicle information is used to identify the cause of problems in the system or undesired operations. Driving information logger can be used in many areas, such as analysis of problems in the study of autonomous driving systems and analysis of accident risk.

2.3. Intelligent Vehicle Path Planning Based on Optimized A* Algorithm

This work by Liang Chu et al explains the rapid development of intelligent driving technology. Achieving accurate path planning for autonomous vehicles has become increasingly crucial. However, path planning algorithms face challenges when dealing with complex and ever-changing road conditions. In this paper, aiming at improving the accuracy and robustness of the generated path, a global programming algorithm based on optimizing is proposed, while maintaining the efficiency of the traditional A* algorithm.

2.4 Research on Autonomous Vehicle Path Planning Algorithm Based on Improved RRT* Algorithm and Artificial Potential Field Method

This work by Xiang Li et al focuses on research on AV path planning algorithms based on improved version of RRT* algorithm and Artificial field method. For the RRT* algorithm, there are problems such as great randomness, longer time consumption, more redundant nodes and inability to perform local obstacle avoidance when encountering unknown obstacles in the path planning process of autonomous vehicles, and the artificial potential field method (APF) applied to autonomous vehicles is prone to problems such as local optimality, unreachable targets and inapplicability to global scenarios.

2.5 Design of an autonomous surface vehicle (ASV) for swarming application

This work by Majid et al focuses on designing ASV for swarming applications. It shows Autonomous surface vehicle is a versatile marine vehicle developed to fulfill complexity requirement of wide range offshore activities. Currently, ASV's are primarily developed based on the conventional surface vessel design in term of body structure and maneuverability capability.

2.6 Real-Time Dynamic Trajectory Planning for Autonomous Driving Vehicles

This work by Wang et al presents real time dynamic trajectory planning algorithm that provides an optimal trajectory along with appropriate acceleration and speed for autonomous driving vehicles. To this end, a center line is first constructed based on a group of predesigned center waypoints that are generally used as a reference line.

2.7 Path planning algorithms in the autonomous driving system

This work by Reda et al focuses on the Autonomous driving system, which aims to reduce human errors, that are the reasons for about 95% of car accidents. The ADS consists of six stages: sensors, perception, localization, assessment, path planning, and control. We explain the main state-of-the-art techniques used in each stage, analyzing 275 papers with 162 specifically on path planning due to its complexity NP-Hard optimization nature, and pivotal role in ADS.

2.8 Research Status of Autonomous Vehicle Path Planning

This work by Shi et al focuses on latest research and applications of path planning algorithms and related hardware for autonomous vehicles. The introduced algorithms include improvement to traditional algorithms, intelligent optimization algorithms, reinforcement learning algorithms and hybrid algorithms such as improving algorithm running speed, path planning ability, and environment learning speed.

2.9 Vehicle Testing Based on Digital Twins Theory for Autonomous Vehicles

This work by Meng et al focuses on vehicle testing based on digital twin theory for autonomous vehicles. Digital twins' theory can provide an effective solution for the complex whole vehicle testing, which includes the dynamics and scenarios. The parallel system is the most advanced method for implementing parallel control theory. In this paper, it introduced virtual systems established by digital modelling the autonomous driving vehicle and scenario, it introduces a connection between virtual system and the real system.

3. OBSERVATION FROM LITERATURE REVIEW

Most research focuses on complex real-world environments, but from our observation we have observed that, developing autonomous vehicle for real world environment is very complex. So, for our requirement for controlled environments, we will be using easier and simpler modular techniques. It is easy to implement autonomous navigation in controlled environment with simpler technique such as A* algorithm for path planning and so on. Our system is designed specifically for controlled environment where the path is predefined for path planning.

In the field of path planning, we observed that A* path planning algorithm is deterministic, efficient and suitable for grid and graph environments. Other algorithms such as RRT/Advanced methods are complex, non-deterministic and are not ideal for structured environments. So, we choose A* for our use case. In our system, one of our use cases defined is dynamic re-planning. For multiple and frequent path planning, the system requires a faster and efficient algorithm. So, the best option was A* algorithm.

From literature review, most of the systems use expensive sensors, require high computation and complex integration. We observed that the systems from the literature review are not practical for retrofitting existing vehicles and low-cost deployment. Our design includes modular system, circuit simulation in the software called TinkerCAD and scalable architecture.

Also from literature review, many systems use digital twins and simulation for testing their system and research. We observed that, simulation is cost-effective, safe and good for validation. In our implementation, Dashboard based simulation, A* testing and dynamic replanning validation are used for testing.

4. LIMITATION OF THE LITERATURE REVIEW

4.1 High system complexity

From literature review, many systems use LiDAR, multi sensor fusion and Advanced AI techniques. But we have observed that it requires high computation and hardware complexity, which is difficult to implement in small scale or retrofit systems. Our system uses simplified architecture and simulation for path planning, obstacle avoidance and dynamic re-planning.

4.2 High cost of deployment

From literature review, we understood that Autonomous system requires expensive hardware, which is not feasible for existing vehicles and small organizations. Our project focuses on low-cost retrofit concept, which requires low-cost alternative hardware for scalability and cost effectiveness.

4.3 Poor suitability for Controlled Environments

From literature review, we understood that most work focuses on urban driving and complex real-world scenarios which is not what we planned for this project. We built our system around controlled environments such as factories, shipyards, airports. Our approach focuses on controlled environments.

4.4 Lack of Real-Time Adaptability

From literature review, we observed that many algorithms focus on static path planning from a start location to goal location. But in real world scenarios, conditions can change so does the destination or the initial path planned by the system. The system requires dynamic obstacle handling and should be able to adjust to unexpected changes. Our take on this issue was to create dynamic re-planning using A* algorithm. At any point in journey, if the user requires or the vehicle comes across a condition where the destination or path must be changed then the system can replan the route from the point it exists.

4.5 Limited Practical Validation

From literature review, we observed that many works are mostly theoretical, simulation-only, or overly complex hardware setups. This lacks simple demonstrable prototypes which can show the validity of the research through practical application. In our project we have created a dashboard simulation which is a prototype for the real dashboard were location, path planning and other computation happens also we have made a circuit design using TinkerCAD which validates the hardware connection. This implementation created validation for our proof-of-concept.

5. PROBLEM DEFINITION AND IMPLEMENTED APPROACH

5.1 Problem Definition

Autonomous navigation technologies have the potential to improve efficiency, safety, and coordination of vehicles operating in controlled environments such as factories, campuses, airports and shipyards. However, converting existing manually operated vehicles into fully autonomous systems is costly and requires complex hardware integration. Although significant improvements and innovations are being made in the field of autonomous driving systems, the adoption of these systems is still in its infancy due to the high costs of the hardware components required for the system's implementation. Most of the existing systems are designed and implemented with the assumption of a completely autonomous platform and are thus difficult to implement in industries and organizations with existing manually driven vehicles.

5.2 Implemented Approach

This project addresses navigation in controlled environment by using A* algorithm for path planning and dynamic replanning of route in unexpected situations or obstacle detection. The system is designed to work in controlled environment with low cost retrofitted vehicles. This will solve the problem of converting all vehicles to autonomous retrofitting the existing vehicles which can have autonomous feature such as path planning, perception for obstacle detection, dynamic replanning and such. The implemented system focuses on path planning in graph based controlled environment using A* algorithm, this is part of our proof-of-concept framework where we have also decided on circuit design for prototype version which can be scaled for real sized vehicles which can be retrofitted.

5.3 Hardware Requirements

Arduino Uno R3

Ultrasonic sensor HCSR04

DC Motor

L293D Motor Driver

Power Supply

Resistors and Capacitors

Servo Motor

Buzzer and LCD (For validation)

5.4 Software Requirements

Python for A* path planning algorithm

HTML and JavaScript for dashboard

Arduino C for hardware logic

TinkerCAD for circuit simulation

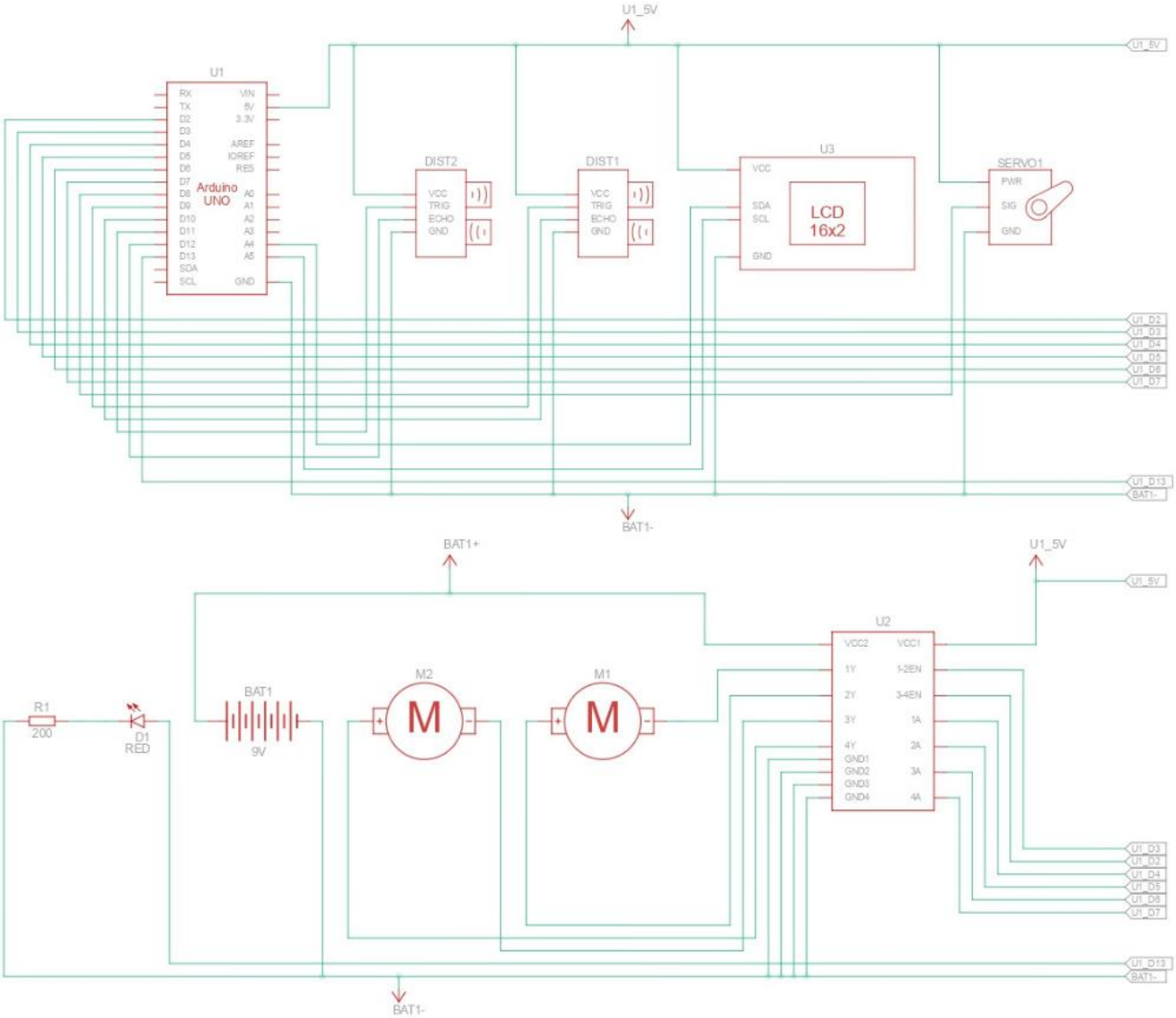

Figure 1: Circuit Design

6. METHODOLOGY

6.1 Identify the needs of customers and users through design thinking methods

Autonomous driving systems have become essential in industries and institutions where efficiency and safety are top priorities. However, most of the existing vehicles are still manually operated, leading to higher manpower costs, operational inefficiency and safety risks. Through different stages of design thinking we identified that the main customer and user needs include:

- Reducing human involvement in repetitive vehicle operations
- Minimizing conversion costs of replacing existing vehicles with autonomous vehicles.\
- Enabling safe, reliable and efficient navigation in controlled environments like factories, shipyards, colleges and so on.
- Providing retrofit solution that can be integrated into existing electric vehicles without major hardware redesign.

By analysing stakeholders such as industrial users, vehicle operators and administrators we concluded that an AI based path finding system capable of autonomous navigation within limited environments would effectively meet these user's requirements.

6.2 Translate the needs of customer into product / process / project

In response to these needs, the proposed project was conceptualized. Our goals as part of this project as in product, process and project are:

Product Goal: To develop an AI module that can guide a vehicle autonomously along a defined route

Process Goal: To design and simulate algorithms, circuits and control logic required for autonomous movement.

Project Goal: To demonstrate a prototype that can later be implemented on a real vehicle user in many industries.

The solution integrates AI pathfinding with hardware control and monitoring systems, forming a hybrid system that demonstrates autonomy though simulation. In the next phase this system will be physically implemented into a life-sized vehicle to validate its effectiveness in a real-world environment.

6.3 Specification of the product / process / project

These specifications define how the proposed system will function and what components will be required to realize it.

Functional Specifications:

The system must detect obstacles and plan an optimal route to the destination using the A* algorithm. It must generate control commands for vehicle movement through a real-time dashboard. The system should be upgraded for integration with any vehicle.

Non-Functional Specifications:

The system should operate safely within a fixed, controlled environment, it should minimize computational cost and maintain real-time response. Must provide an emergency stop feature and manual override in real prototype.

6.4 Designing the solution for required customed needs

The methodology adopted for the project presents the steps for a systematic approach in designing, simulating, and implementing an AI-based pathfinding system that could be integrated into a non-autonomous electric vehicle. The approach merges hardware design, algorithm development, and simulation testing into one comprehensive package that is scalable and cost-effective for autonomous navigation within a controlled environment, such as a college campus or an industrial site.

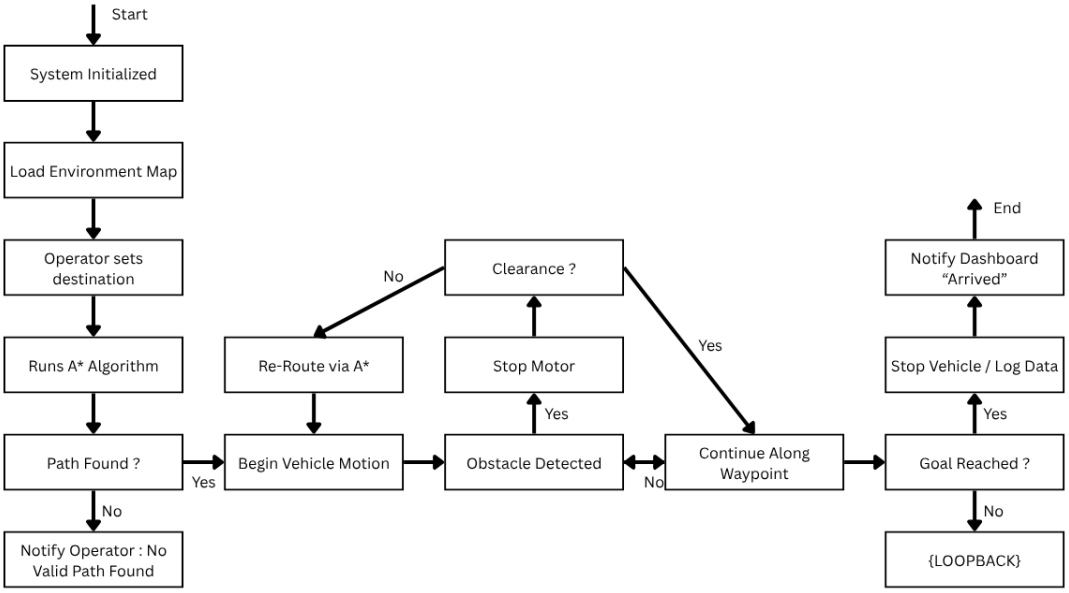

Figure 2: Activity Diagram

7. MODULE DESCRIPTION

7.1 Input and Goal Management Module

This module takes destination from user as input, it is then converted into coordinates, it will pass the information to other modules after it is validated. This is the starting point of system.

7.2 Perception Module

This module is one of the most important parts of the project, it uses simulated sensor data. It contains Ultrasonic sensor which senses distance between vehicle and object in the range. This module detects obstacles and provides environment data. Simulated sensor inputs are used for validation.

7.3 Path Planning Module

This module is also one of the most important parts of the project, it uses A* algorithm to calculate the path from the current position to the destination. It calculates the shortest path for the destination using A* path planning algorithm. It is also responsible for dynamic replanning which recalculates path after stop/obstacle.

7.4 Navigation and Control Module

This module is responsible for converting path into movement, it generates waypoints and control signals.

7.5 Dashboard/Monitoring Module

This module is responsible for visual interface of dashboard, it shows map, path, vehicle position in the map and vehicle status. It includes real time updates of the vehicle and replanning visualization.

7.6 Hardware Interface Module

This module connects software with hardware, this includes controller board, motor driver, sensors and hardware component. This is designed and validated using TinkerCAD.

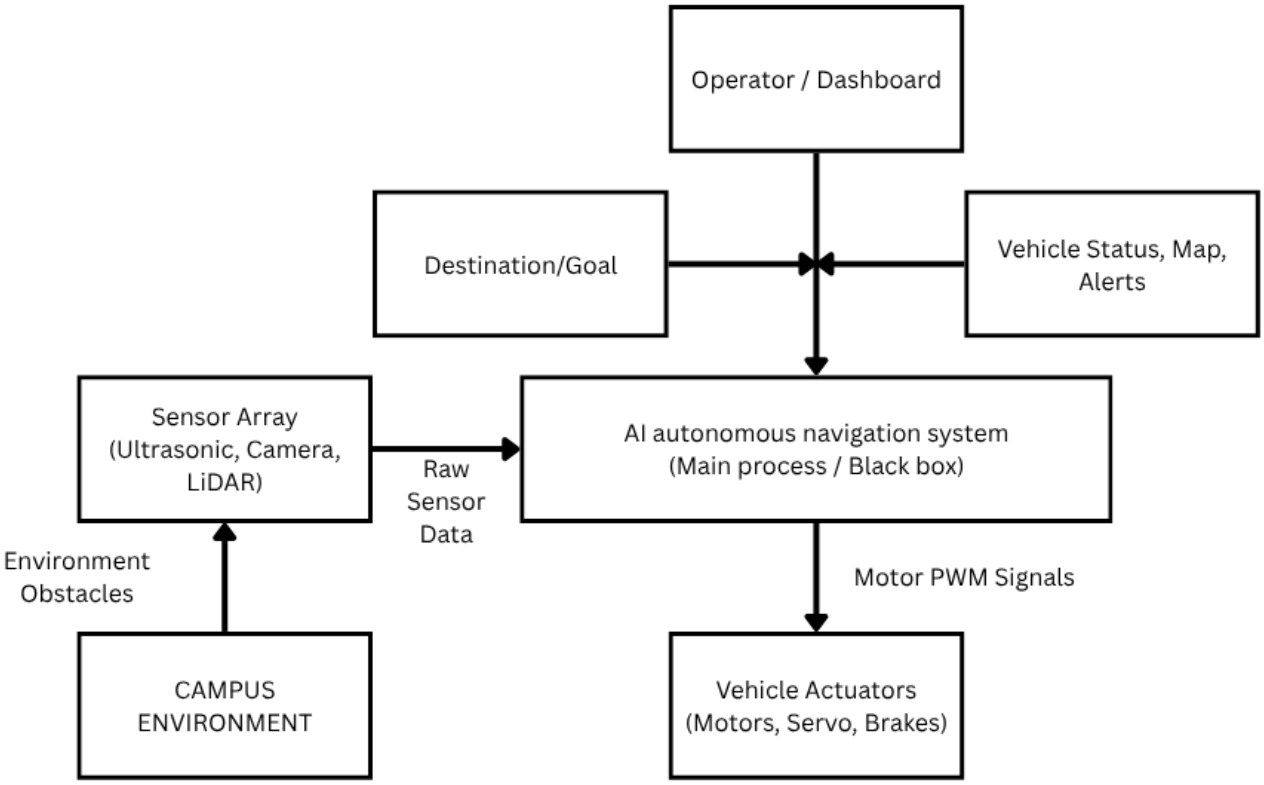

Fig 3: Data Flow Diagram

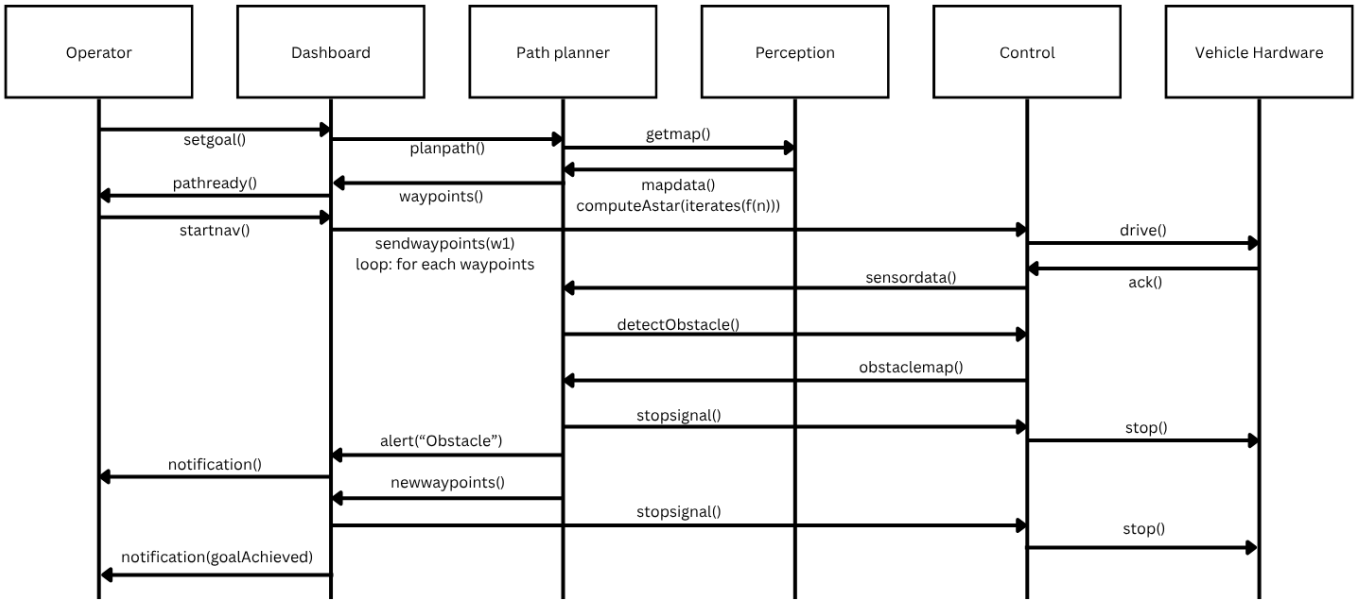

Fig 4: Sequence Diagram

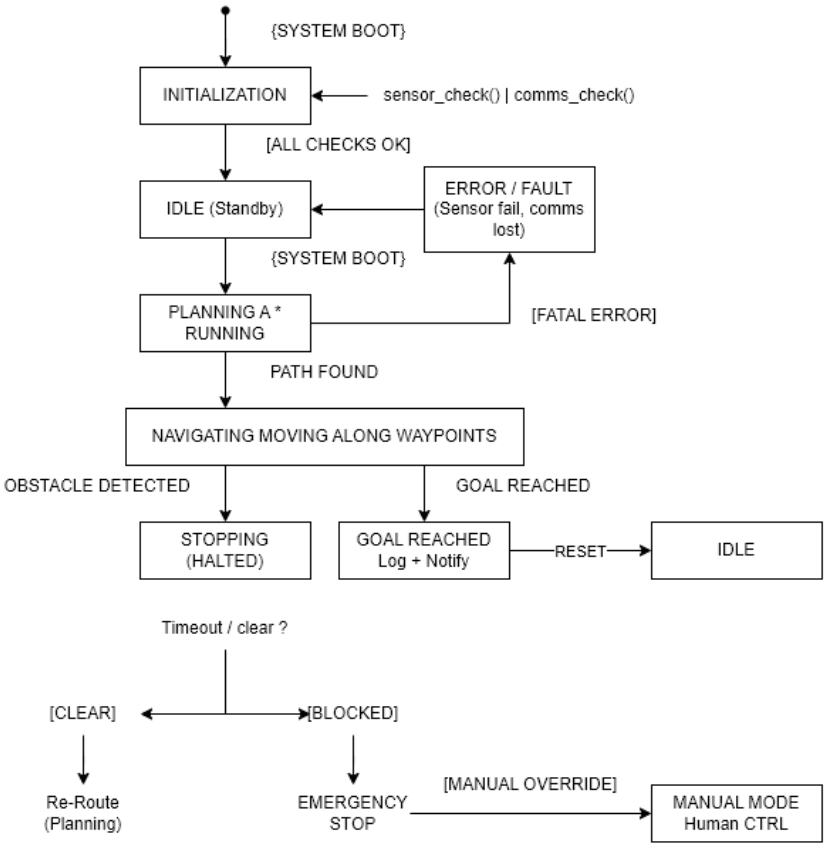

Fig 5: State Diagram

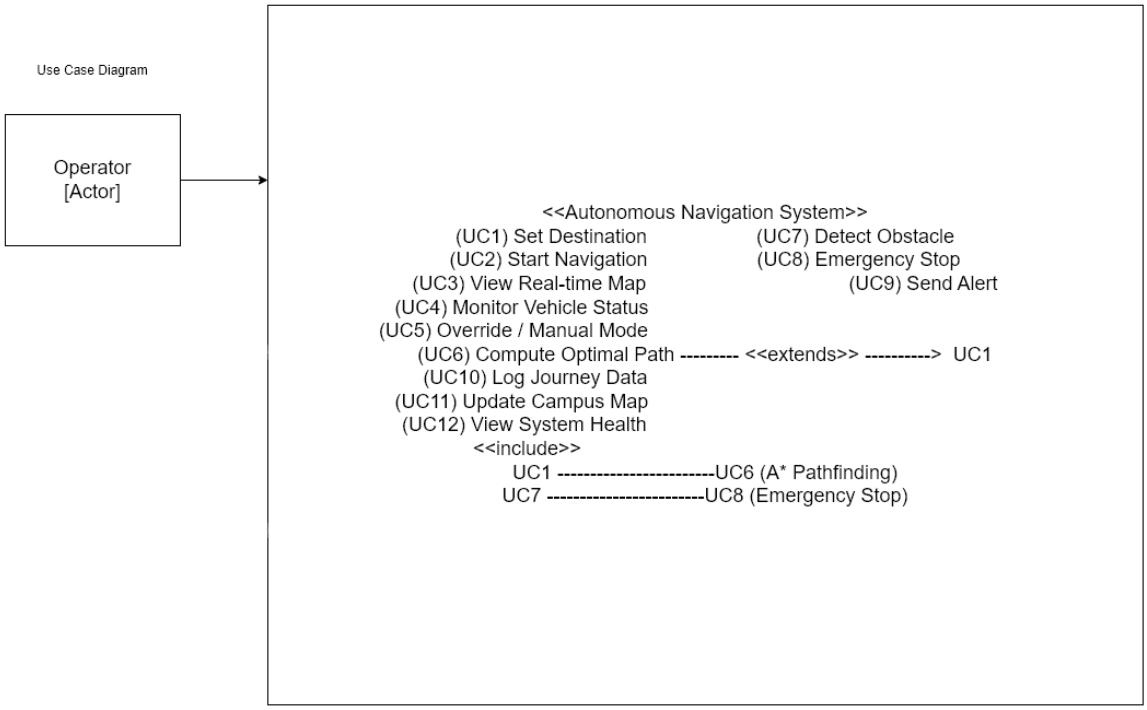

Fig 6: Use case Diagram

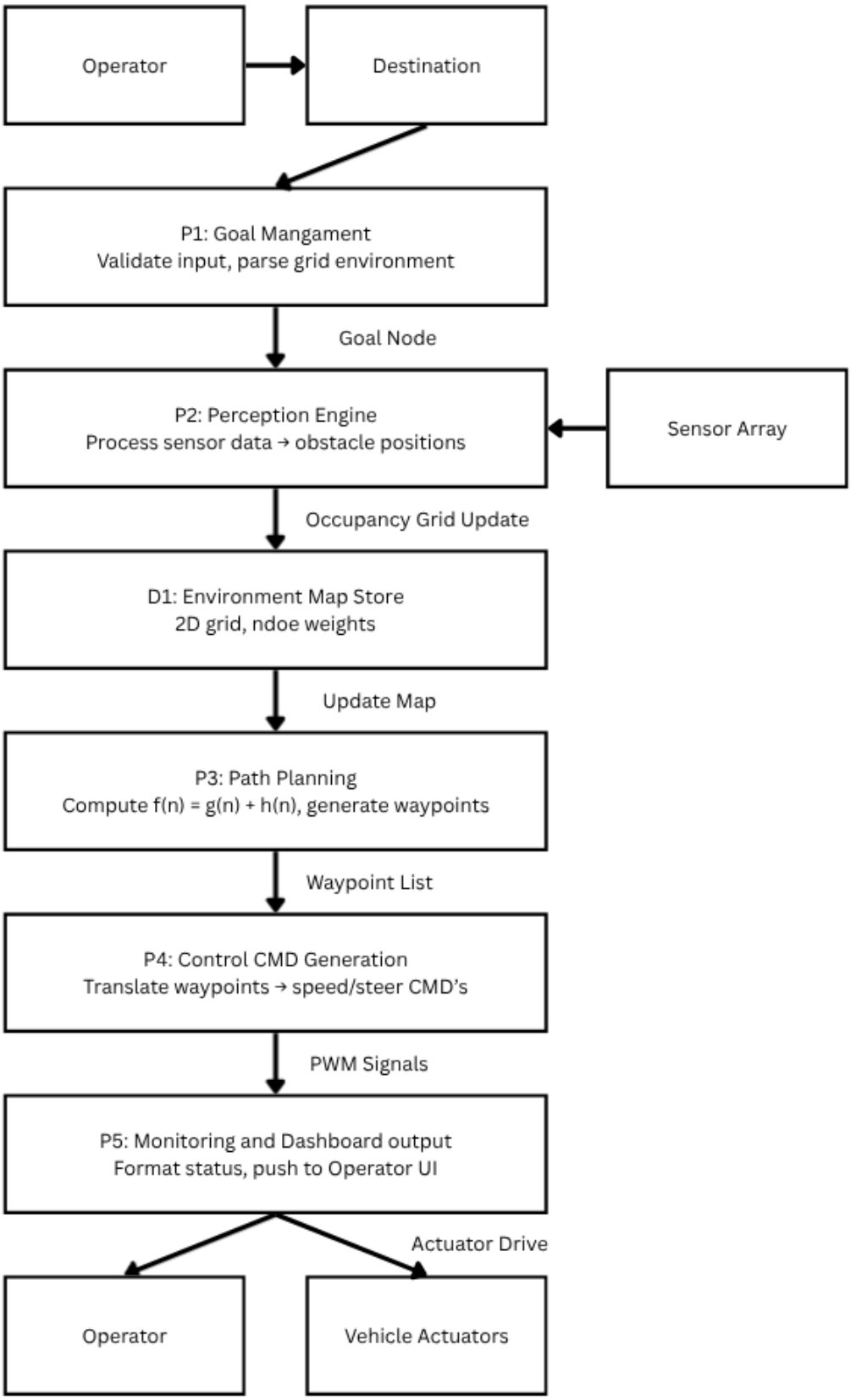

Fig 7: Data Flow Diagram as per phases

8. EXPERIMENTATION

8.1 Experimental Setup

We have created a dashboard prototype using HTML and JavaScript to test A* algorithm on a custom map which we have created. The dashboard prototype is a webpage which represent how the dashboard should work in real scenario. The user can give start and goal location from few options given in the dashboard, in the prototype as a visual demonstration a vehicle will travel through the calculated path and then shows all the metrics such as speed, distance travel, etc.

The dashboard prototype is saved in GitHub at: tinyurl.com/dp-dashboard-prototype

8.2 Testing and Metrics

The dashboard prototype is a proof of concept of the proposed system which shows that in controlled environments, A* algorithm is the best path planning method for navigating autonomous vehicles. In the dashboard, users have few option to select as start and goal location of the given map. After selecting the location, user can start navigation, which will signal the path planning module to calculate the route. Then the vehicle will start moving through the selected path, there is emergency stop button and manual override button which can be used for required scenarios thus making the vehicle also work without autonomous feature or navigation system. The dashboard also logs all the information when navigation start which can be seen in system log menu. In our testing, the system was able to find optimal path towards the goal location all the times making it 100% working for navigation system. As we have mentioned this is a proof of concept for the framework. For the testing purposes, the system is evaluated based on:

1. Path length: Total distance between start and goal position.
2. Computation Time: Time required to calculate the path using the A* algorithm
3. Replanning Frequency: Number of times the system triggers path replanning due to dynamic obstacles.

8.3 Tools and Implementation

The dashboard is made using HTML and JavaScript, dashboard shows the map in the centre and options for controlling the dashboard are given on left sidebar, the right side contains all the vitals of the vehicle. It also contains a system log menu which will log all the details of the vehicle, navigation, etc.

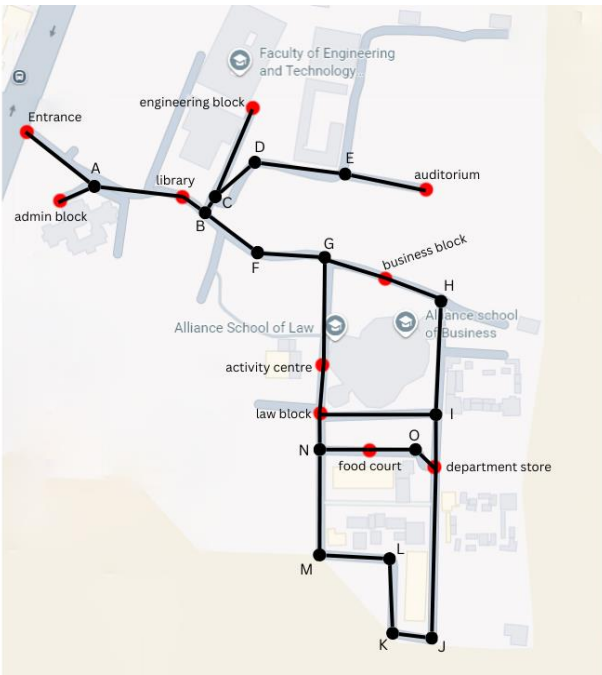

Fig 8: Custom Map

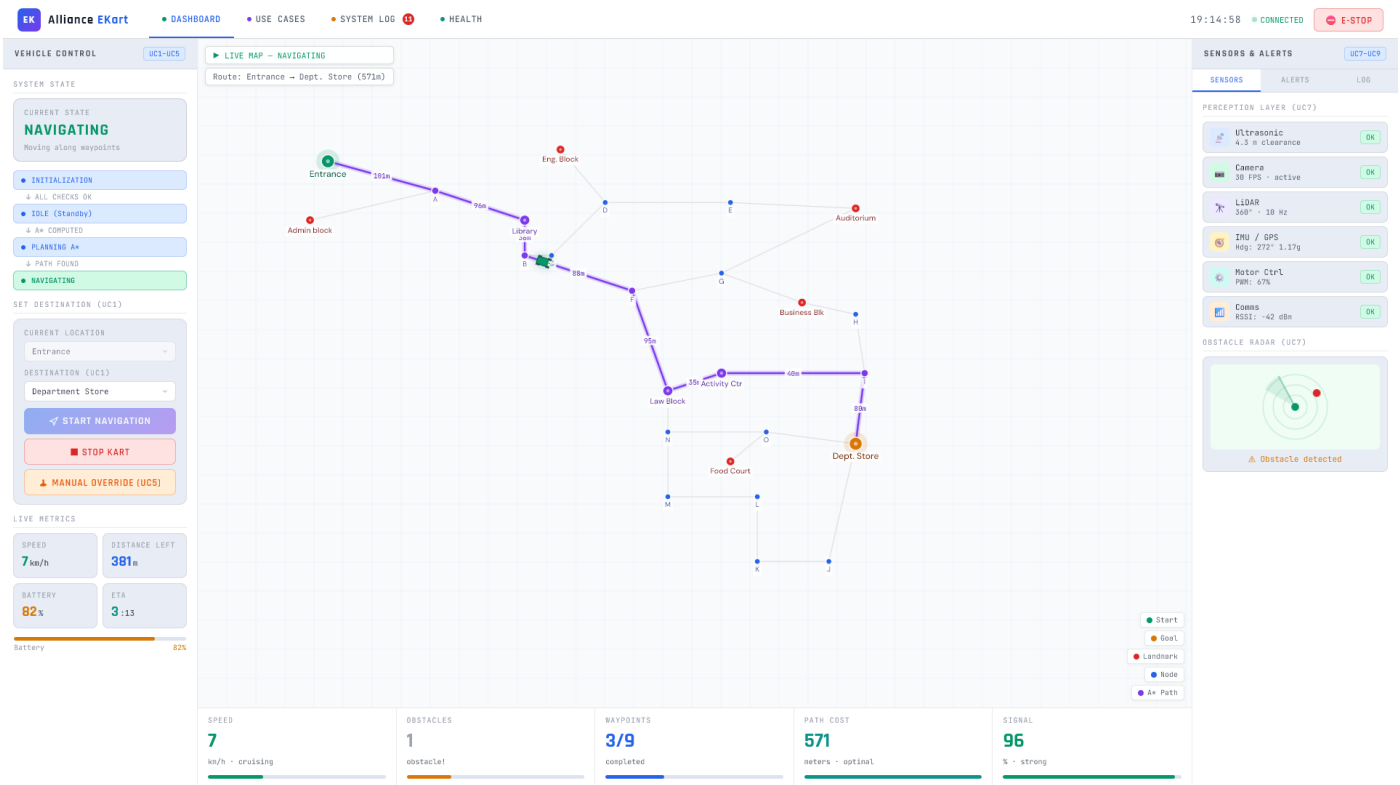

Fig 9: Dashboard Prototype

9. RESULT AND DISCUSSION

9.1 Results

In our testing, we observed that A* algorithm always found valid path, optimality is also maintained. The trend in the testing results showed that:

- More nodes require more computation time
- More obstacles/replans result in longer path
- Dynamic obstacles result in higher replanning

Based on our evaluation across multiple iterations:

Path length	Average Distance (meter)	Average path planning time (ms)	Average journey time (sec)
Very small	110m	12ms	244
small	174m	15ms	386
medium	212m	19ms	471
large	273m	23ms	606
Very large	335m	27ms	744

Table 1: Simulation Metrics

9.2 System behaviour and Efficiency

Also, in our testing we observed that A* algorithm is the fastest path planning algorithm in controlled environments because of the characterism known as Heuristics. The system is stable with any combination of input; it can efficiently determine and adapt to the unseen data by using A* algorithm on the given search space. Thus, making this system efficient and scalable in any search space or map.

9.3 Limitations

In our observation we also found some limitation of this system. The system only works well in controlled environments so it can't be used in a dynamic environment such as public roads. Even in controlled environment, the number of nodes and length of edges affect the time of calculating the route. Also, in our testing we didn't consider real world noise which affect in unpredicted ways.

10. CONCLUSION AND FUTURE DIRECTION

10.1 Conclusion

This work shows a proof of concept of a simulated system of autonomous navigation in a controlled environment. This system combines the concepts of grid-based environment modelling, perception abstraction, and the A* algorithm of path planning to create efficient paths of navigation. Contrary to existing methods of path planning and navigation that heavily depend on hardware and a large environment of deployment, this study proposes a lightweight and software-based approach to the problem of path planning and navigation in a structured environment. Simulation results of the study prove the effectiveness of classical methods of path planning in the creation of adaptive paths of navigation with minimal computations required. This proposed system provides a strong platform on which further research can be conducted on the implementation of the system in the real world.

10.2 Future Works

As mentioned earlier, we are demonstrating the proof of concept of the proposed framework. Currently we are making this project Open Source for research, academia and industry. In the future we are expecting to build a new version which includes Real world integration with real sized vehicle, Multi Agent Navigation and Machine Learning Integration.

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